```
import pandas as pd
import numpy as np
from sklearn.datasets import load_breast_cancer
from sklearn.feature_selection import chi2
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Lasso
import statsmodels.api as sm
```

```
# 1. Load Real Dataset
cancer = load_breast_cancer()
X = pd.DataFrame(cancer.data, columns=cancer.feature_names)
y = pd.Series(cancer.target, name='Target')
data = pd.concat([X, y], axis=1)

print(f"Dataset loaded with {X.shape[1]} features.\n")
```

```
Dataset loaded with 30 features.
```

```
# --- TECHNIQUE 1: Correlation Analysis ---
correlation = data.corr()['Target'].abs().sort_values(ascending=False)
print("Top 5 Correlated Features:")
print(correlation[1:6])  # Excluding target itself
print("-" * 30)
```

```
Top 5 Correlated Features:
worst concave points    0.793566
worst perimeter         0.782914
mean concave points     0.776614
worst radius            0.776454
mean perimeter          0.742636
Name: Target, dtype: float64
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```

```
# --- TECHNIQUE 2: Chi-Square Test ---
# Chi2 requires non-negative values
chi_scores, p_values = chi2(X, y)
chi_df = pd.DataFrame({'Feature': X.columns, 'Chi2 Score': chi_scores, 'p-value': p_values})
print("Top 5 Chi-Square Features:")
print(chi_df.sort_values(by='Chi2 Score', ascending=False).head(5))
print("-" * 30)
```

```
Top 5 Chi-Square Features:
            Feature     Chi2 Score  p-value
23       worst area  112598.431564      0.0
3         mean area   53991.655924      0.0
13       area error    8758.504705      0.0
22  worst perimeter    3665.035416      0.0
2    mean perimeter    2011.102864      0.0
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```

```
# --- TECHNIQUE 3: Forward Selection (Wrapper Method) ---
def forward_selection(X, y):
    selected = []
    remaining = list(X.columns)
    while len(selected) < 5:  # Limit to top 5 for brevity
        scores = []
        for feature in remaining:
            model = sm.OLS(y, sm.add_constant(X[selected + [feature]])).fit()
            scores.append((model.rsquared, feature))
        best_score, best_feature = max(scores)
        selected.append(best_feature)
        remaining.remove(best_feature)
    return selected

print("Forward Selection (Top 5):")
print(forward_selection(X, y))
print("-" * 30)
```

```
Forward Selection (Top 5):
['worst concave points', 'worst radius', 'worst texture', 'worst area', 'smoothness error']
------------------------------
```

```
# --- TECHNIQUE 4: Backward Elimination (Wrapper Method) ---
def backward_elimination(X, y, threshold=0.05):
    X_const = sm.add_constant(X)
    while True:
        model = sm.OLS(y, X_const).fit()
        max_p = model.pvalues.max()
```

```
        if max_p > threshold:
            remove_feature = model.pvalues.idxmax()
            X_const = X_const.drop(columns=[remove_feature])
        else:
            break
    return X_const.columns

selected_columns = backward_elimination(X, y)
print("Backward Elimination remaining features count:", len(selected_columns))
print("Selected Feature Names:")
print(list(selected_columns))
print("-" * 30)
```

```
Backward Elimination remaining features count: 14
Selected Feature Names:
['const', 'mean radius', 'mean compactness', 'mean concave points', 'radius error', 'smoothness error', 'concavity error', '
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```

```
# --- TECHNIQUE 5: Lasso Regression (Embedded Method) ---
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
lasso = Lasso(alpha=0.01)  # alpha controls sparsity
lasso.fit(X_scaled, y)

lasso_df = pd.DataFrame({'Feature': X.columns, 'Coefficient': lasso.coef_})
important_features = lasso_df[lasso_df['Coefficient'] != 0]
print(f"Lasso kept {len(important_features)} features. Top ones:")
print(important_features.sort_values(by='Coefficient', key=abs, ascending=False))
print(important_features.sort_values(by='Coefficient', key=abs, ascending=False).head(5))
```

```
Lasso kept 12 features. Top ones:
                   Feature  Coefficient
20            worst radius    -0.166596
27    worst concave points    -0.126911
21           worst texture    -0.051017
7      mean concave points    -0.045709
28          worst symmetry    -0.040223
24        worst smoothness    -0.032805
1             mean texture    -0.019530
10            radius error    -0.018349
9   mean fractal dimension     0.015021
14        smoothness error    -0.009790
26         worst concavity    -0.008220
16         concavity error     0.003305
                  Feature  Coefficient
20          worst radius    -0.166596
27  worst concave points    -0.126911
21         worst texture    -0.051017
7    mean concave points    -0.045709
28        worst symmetry    -0.040223
```

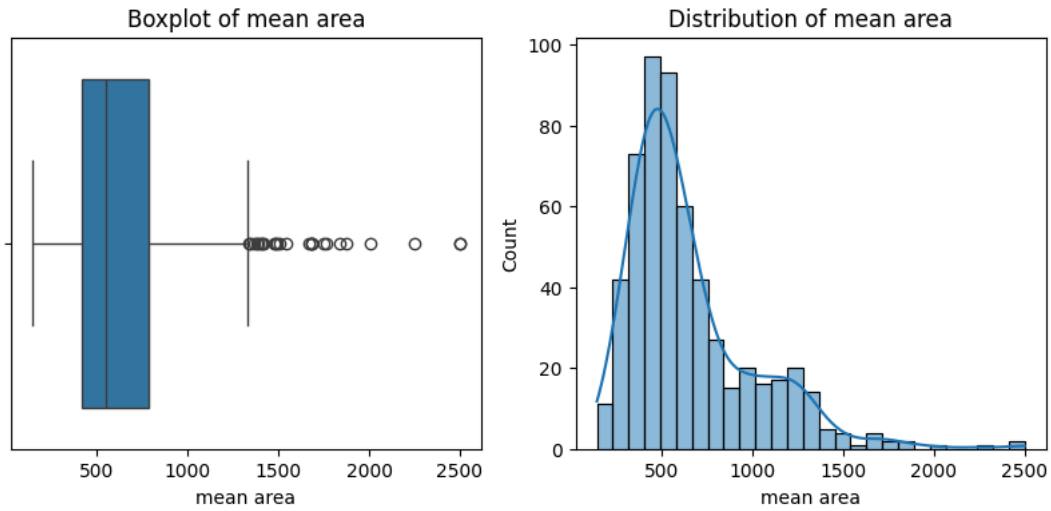
```
import numpy as np
import pandas as pd
from scipy import stats

def detect_outliers_zscore(df, column, threshold=3):
    z_scores = np.abs(stats.zscore(df[column]))
    outliers = df[z_scores > threshold]
    print(f"Z-Score Method: Found {len(outliers)} outliers in '{column}'")
    return outliers

# Example usage with Breast Cancer 'mean area'
outliers_z = detect_outliers_zscore(X, 'mean area')
```

```
Z-Score Method: Found 8 outliers in 'mean area'
```

```
def detect_outliers_iqr(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[column] < lower_bound) | (df[column] > upper_bound)]
    print(f"IQR Method: Found {len(outliers)} outliers in '{column}'")
    print(f"Bounds: [{lower_bound:.2f}, {upper_bound:.2f}]")
    return outliers

# Example usage
outliers_iqr = detect_outliers_iqr(X, 'mean area')
```

```
IQR Method: Found 25 outliers in 'mean area'
Bounds: [-123.30, 1326.30]
```

```
import matplotlib.pyplot as plt
import seaborn as sns

def plot_outliers(df, column):
    plt.figure(figsize=(10, 4))

    # Subplot 1: Boxplot
    plt.subplot(1, 2, 1)
    sns.boxplot(x=df[column])
    plt.title(f'Boxplot of {column}')

    # Subplot 2: Histogram
    plt.subplot(1, 2, 2)
    sns.histplot(df[column], kde=True)
    plt.title(f'Distribution of {column}')

    plt.show()

plot_outliers(X, 'mean area')
```

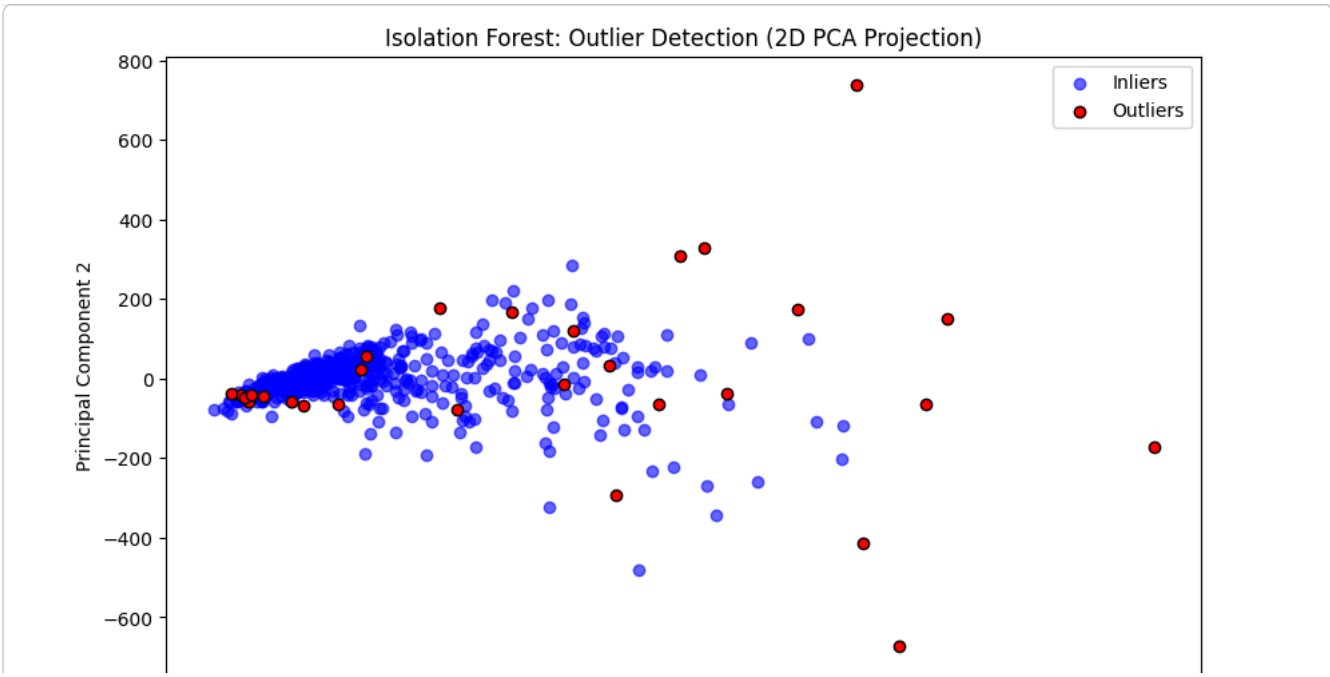



```
import pandas as pd
from sklearn.ensemble import IsolationForest
from sklearn.datasets import load_breast_cancer

# 1. Load your dataset
cancer = load_breast_cancer()
df = pd.DataFrame(cancer.data, columns=cancer.feature_names)
print(f"Original dataset shape: {df.shape}")

# 2. Initialize Isolation Forest
# contamination: the expected proportion of outliers (e.g., 0.05 = 5%)
iso_forest = IsolationForest(n_estimators=100, contamination=0.05, random_state=42)

# 3. Fit and Predict
# Returns 1 for inliers and -1 for outliers
df['anomaly_score'] = iso_forest.fit_predict(df)

# 4. Separate Outliers and Inliers
outliers = df[df['anomaly_score'] == -1]
clean_df = df[df['anomaly_score'] == 1].drop(columns=['anomaly_score'])

print(f"Detected outliers: {len(outliers)}")
print(f"Cleaned dataset shape: {clean_df.shape}")

# 5. (Optional) View the features of the most "extreme" outliers
print("\nSample of detected outliers:")
print(outliers.head())
```

```
Original dataset shape: (569, 30)
Detected outliers: 29
Cleaned dataset shape: (540, 30)

Sample of detected outliers:
    mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0         17.99         10.38          122.80     1001.0          0.11840
3         11.42         20.38           77.58      386.1          0.14250
9         12.46         24.04           83.97      475.9          0.11860
```

```
12        19.17         24.80          132.40     1123.0          0.09740
42        19.07         24.81          128.30     1104.0          0.09081

    mean compactness  mean concavity  mean concave points  mean symmetry  \
0             0.2776          0.3001              0.14710         0.2419
3             0.2839          0.2414              0.10520         0.2597
9             0.2396          0.2273              0.08543         0.2030
12            0.2458          0.2065              0.11180         0.2397
42            0.2190          0.2107              0.09961         0.2310

    mean fractal dimension  ...  worst texture  worst perimeter  worst area  \
0                  0.07871  ...          17.33           184.60      2019.0
3                  0.09744  ...          26.50            98.87       567.7
9                  0.08243  ...          40.68            97.65       711.4
12                 0.07800  ...          29.94           151.70      1332.0
42                 0.06343  ...          33.17           177.40      1651.0

    worst smoothness  worst compactness  worst concavity  \
0             0.1622             0.6656           0.7119
3             0.2098             0.8663           0.6869
9             0.1853             1.0580           1.1050
12            0.1037             0.3903           0.3639
42            0.1247             0.7444           0.7242

    worst concave points  worst symmetry  worst fractal dimension  \
0                 0.2654          0.4601                   0.1189
3                 0.2575          0.6638                   0.1730
9                 0.2210          0.4366                   0.2075
12                0.1767          0.3176                   0.1023
42                0.2493          0.4670                   0.1038

    anomaly_score
0              -1
3              -1
9              -1
12             -1
42             -1

[5 rows x 31 columns]
```

```
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.ensemble import IsolationForest

# 1. Reduce 30 features to 2 for 2D plotting
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

# 2. Fit Isolation Forest on the reduced data
iso_forest = IsolationForest(contamination=0.05, random_state=42)
preds = iso_forest.fit_predict(X)  # Fitting on full 30 features for accuracy

# 3. Create the Visualization
plt.figure(figsize=(10, 6))
plt.scatter(X_pca[preds == 1, 0], X_pca[preds == 1, 1], c='blue', label='Inliers', alpha=0.6)
plt.scatter(X_pca[preds == -1, 0], X_pca[preds == -1, 1], c='red', label='Outliers', edgecolors='k')

plt.title("Isolation Forest: Outlier Detection (2D PCA Projection)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.show()
```


Isolation Forest: Outlier Detection (2D PCA Projection)
Inliers
Outliers
Principal Component 2
800
600
400
200
0
−200
−400
−600